Muhammad Muneeb

Software Engineering Student | Python & Data Analytics

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PROFESSIONAL SUMMARY

Software Engineering student with hands-on experience in Python programming, data analysis, feature engineering, and machine learning projects. Currently completing an AI/ML internship at NETSOL Technologies Pakistan through the NIAI training program. Developed projects in text classification, loan approval prediction, audio anomaly detection, API-based chatbots, and data pipelines. Familiar with the end-to-end workflow from data preparation and model evaluation to prediction, with foundational knowledge of Azure, RAG, workflow automation, and MLOps.

TECHNICAL SKILLS

Programming and Tools: Python, SQL basics, object-oriented programming, Git, GitHub, VS Code, Jupyter Notebook, Streamlit

Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Seaborn, exploratory data analysis (EDA), descriptive statistics, data profiling

Data Preparation: Data cleaning, preprocessing, missing-value imputation, IQR outlier detection, feature engineering, feature selection, ordinal encoding, label encoding, one-hot encoding, scaling, normalization

Machine Learning: Scikit-learn, supervised and unsupervised learning, classification, regression, model training, train/test split, cross-validation, hyperparameter tuning, loss functions, prediction, overfitting and underfitting

Algorithms: Linear Regression, Polynomial Regression, Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Naive Bayes, SGDClassifier, Isolation Forest

Model Evaluation: Accuracy, precision, recall, F1 score, R-squared, mean absolute error (MAE), root mean squared error (RMSE), model comparison and performance analysis

NLP and Audio: Basic natural language processing, text classification, TF-IDF, MFCC feature extraction, librosa, sounddevice, soundfile

Data Engineering: ETL pipelines, REST API integration, requests, JSON, SQLite, database integration, structured data processing, joblib model persistence

Cloud and AI Foundations: Microsoft Azure, Retrieval-Augmented Generation (RAG), n8n workflows, AI agents, MLOps concepts

EXPERIENCE

AI/ML Internship | NETSOL Technologies Pakistan

October 2026 - Present | Lahore, Pakistan

- Selected for a two-week AI/ML internship through the NIAI training program based on performance.
- Working on practical AI/ML tasks, including a Streamlit chatbot using the Grok API, data handling, and machine learning workflows.

EDUCATION AND TRAINING

BS Software Engineering | 2023–2027 (Expected) The Islamia University of Bahawalpur

AI/ML Training | NETSOL NIAI / NAVTTC September 2026 - Present

Practical training in Python, data analysis, machine learning, ETL, NLP, and Azure AI services through exercises and projects.

PROJECTS

MachineSense AI | Audio Anomaly Detection Prototype

Python, Scikit-learn, librosa, Isolation Forest, StandardScaler, joblib

- Collaborating on a predictive maintenance project designed to learn normal machine behavior and detect unusual patterns for small and medium-sized businesses.
- Built an audio-only prototype using motor recordings, 40 MFCC features, feature scaling, and an Isolation Forest model to identify changes in machine sound.
- Separated training and calibration segments and used a calibration-based anomaly threshold for live microphone testing. Vibration, temperature, current, and RPM integration are planned extensions.

Email Spam Detector | NLP Classification

Python, Pandas, TF-IDF, Scikit-learn

- Prepared approximately 480 cleaned email records and combined TF-IDF text features with subject length, body length, attachment status, and link counts.
- Compared Logistic Regression, Naive Bayes, Random Forest, KNN, Decision Tree, and SGD classifiers on a train/test split.
- Achieved 97.92% accuracy, 100% precision, 95.24% recall, and 97.56% F1 score with Logistic Regression on a 96-record test set.

Loan Approval Predictor | Tabular Classification

Python, Pandas, Scikit-learn, Decision Tree

- Prepared financial and applicant data, encoded education and self-employment features, and trained a classification model to predict loan approval outcomes.
- Achieved approximately 97% test accuracy with a Decision Tree and added input-based prediction using the trained model.

Grok API Chatbot | Streamlit Application

Python, Streamlit, OpenAI-compatible API client

- Developed a chat interface with session-based conversation history, a password-masked API key field, configurable model input, and a clear-chat control.
- Integrated Grok through its API and added API error handling to make request failures visible in the application.

Product Data ETL Pipeline | API to SQLite

Python, requests, Pandas, SQLite, JSON

- Extracted product data from the DummyJSON API, transformed records using Pandas, and loaded the results into a SQLite database.
- Handled missing values, renamed fields, rounded numeric values, and read stored records back into a DataFrame for validation and analysis.

Freight Rate Prediction | Machine Learning Workflow

Python, Pandas, Scikit-learn, Jupyter Notebook

- Built a freight-rate prediction workflow covering data preprocessing, exploratory analysis, feature engineering, train/test splitting, model training, evaluation, and prediction.

PROFESSIONAL SKILLS

Problem solving, analytical thinking, team collaboration, technical documentation, communication, and continuous learning.